

Geometry Basics

SUPPORTING SKILLS LEGEND

Student learning objectives for this unit

I — Introduce

D — Deepen

A — Apply

R — Reinforce

- ☐ 1. I can **name and represent** points, lines, planes, segments, and rays using correct notation.

RELATED SKILLS

- I Add congruence, parallel, perpendicular, and angle markings to a diagram.
- I Read and write geometric notation correctly.
- I Recall definitions of points, lines, planes, segments, rays, angles, and intersections.

- ☐ 2. I can **identify** collinear, coplanar, intersecting, and concurrent figures.

RELATED SKILLS

- I Distinguish stated or marked facts from visual appearance.
- A Read and write geometric notation correctly.

- ☐ 3. I can **calculate** segment lengths using betweenness and the segment-addition postulate.

RELATED SKILLS

- I Translate segment addition into an equation.
- A Translate quantities and relationships into an equation.
- A Use inverse operations to maintain an equivalent equation.

- ☐ 4. I can **determine** a midpoint or use a segment bisector relationship.

RELATED SKILLS

- I Use a bisector to create equal measures.
- I Use a midpoint to create equal segment lengths.
- A Translate segment addition into an equation.

- ☐ 5. I can **classify and measure** angles.

RELATED SKILLS

- I Identify complementary, supplementary, vertical, and linear-pair relationships.
- I Measure or calculate an angle accurately.

- ☐ 6. I can **calculate** angle measures using angle addition and angle bisectors.

RELATED SKILLS

- I Translate angle addition into an equation.
- A Use a bisector to create equal measures.
- A Use inverse operations to maintain an equivalent equation.

☐ **7. I can **identify and use** complementary, supplementary, vertical, and linear-pair relationships.**

RELATED SKILLS

- ☐ D Identify complementary, supplementary, vertical, and linear-pair relationships.
 - ☐ A Translate quantities and relationships into an equation.
 - ☐ A Use inverse operations to maintain an equivalent equation.
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☐ **8. I can **distinguish** a definition, postulate, theorem, converse, and counterexample.**

RELATED SKILLS

- ☐ I Apply a geometric definition as justification.
 - ☐ I Disprove a universal statement with a counterexample.
 - ☐ I Recognize and apply the converse of a conditional statement.
 - ☐ I Select a theorem that connects known facts to a target statement.
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☐ **9. I can **determine** angle relationships formed by parallel lines and a transversal.**

RELATED SKILLS

- ☐ I Identify corresponding, alternate, and same-side angle pairs.
 - ☐ I Use parallel-line angle relationships.
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☐ **10. I can **use** angle relationships to determine whether lines are parallel.**

RELATED SKILLS

- ☐ I Use angle relationships to prove lines parallel.
 - ☐ A Identify corresponding, alternate, and same-side angle pairs.
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☐ **11. I can **solve** problems involving parallel, perpendicular, vertical, or horizontal lines.**

RELATED SKILLS

- ☐ I Use right-angle relationships created by perpendicular lines.
 - ☐ A Use inverse operations to maintain an equivalent equation.
 - ☐ A Use parallel-line angle relationships.
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☐ **12. I can **construct** a logical geometric argument from stated facts and accepted results.**

RELATED SKILLS

- ☐ I Organize statements and reasons into a valid proof sequence.
 - ☐ I Translate givens into diagram markings and statements.
 - ☐ A Select a theorem that connects known facts to a target statement.
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